Lecture 20 – Artificial Life and Cellular Automata

ECM3412/ECMM409– Nature-Inspired Computation

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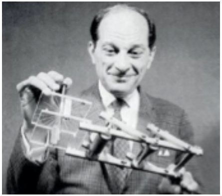
Cellular automata introduction

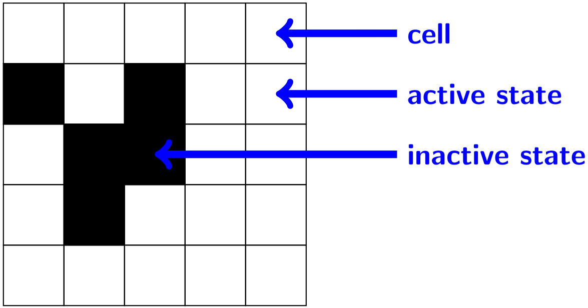
- Originally devised in the late 1940s by Stan Ulam (a mathematician) and John von Neumann
- Intended as a method of representing a stylised universe with rules (e.g. thermodynamics) acting over the entire universe
- In an A-life context – a demonstration of convergence: complex global behaviour emerges from local interactions following simple rules
- Have subsequently been used for a wide variety of purposes in simulating systems from chemistry, physics and biology



Stan Ulam

John von Neumann

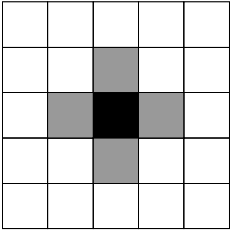
Cellular automata



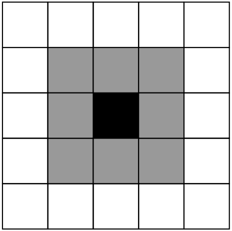
- An automaton consists of a grid of cells each of which can be in a (small and finite, normally binary) number of states
- The figure shows a 5×5 automaton where each cell can be active or inactive

Neighbourhoods

- Each cell has a neighbourhood
- Neighbourhoods determine the extent of interaction between cells in the grid
- Two popular neighbourhoods are the **von Neumann** and **Moore** neighbourhoods



von Neumann



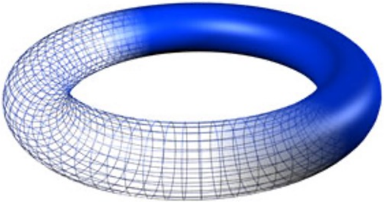
Moore

Topology

An automaton can be

- **1D** – i.e. a line of cells
- **2D** – as we've already seen
- **3D+** – theoretically there is no limit to dimensionality

Automata are often toroidal – cells 'wrap around' to the other side

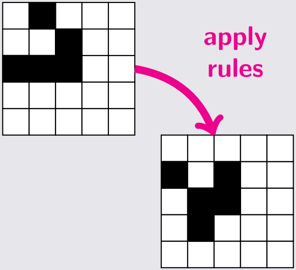


State transition rules

- The state of an automaton changes over time in **discrete timesteps**
- The state of each cell is modified in parallel at each timestep according to **state transition rules**
- These determine the new states of the cells in the next timestep from the states of that cell's neighbours

Execution

- The CA runs by changing the state of the cells according to state transition rules
- The rules depend on the state of the cell and that of its neighbours
- Every cell in the automaton has its rules applied **before** the automaton is updated
- Each timestep the automaton can be seen as a system configuration for that timestep



The 256 elementary CA

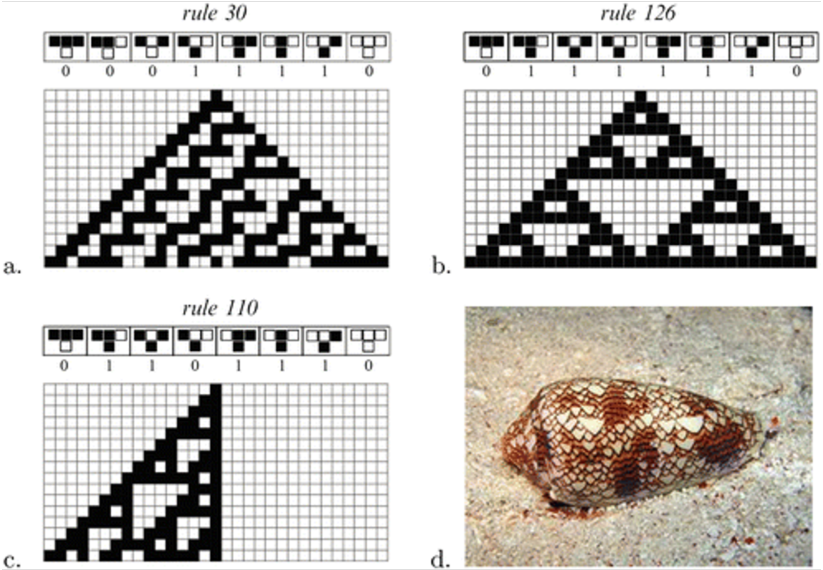
The simplest class of one-dimensional CA proposed by Wolfram

- Binary representation
- Rules depend only on nearest neighbour values
- $2 \times 2 \times 2 = 2^3 = 8$ possible binary states for the three neighbours

$\mathbf{t = 1}$

$\mathbf{t = 2}$

Modelling patterns in nature



Conway's Game of Life

Probably the most well known CA

Rules

1. If a cell is off and exactly three of its neighbours are on then that cell becomes on in the next timestep – otherwise it remains off
2. If a cell is on and has either two or three active neighbours then it remains on – otherwise it becomes off

Even simple rules like this can have unexpected results

https://playgameoflife.com/

Cellular autonoma

Important properties of CAs

- **Localism** – states are updated based on neighbourhood properties
- **Parallelism** – the state of every cell is updated in parallel
- **Homogeneity** – the same set of rules is applied across the automaton

Types of CA

It is not possible to predict in advance what behaviour will be displayed by the CA given its rules – the CA can descend into a number of states

Wolfram proposed a classification scheme based on these criteria:

1. Evolution leads to a homogeneous state
2. Evolution leads to a set of separated simple stable/ periodic structures
3. Evolution leads to a chaotic pattern
4. Evolution leads to complex local structures – sometimes long-lived

Stephen Wolfram, A New Kind of Science https://www.wolframscience.com/nks/

Modelling Covid-19

- Build a CA based on **infection** (confirmed, unconfirmed, asymptomatic) and their **situation** (self-isolating, hospitalised. . .) and the **outcome** of their case
- Transition rule based on a **probability** based on multiple factors
- Case studies based on New York City and Iowa

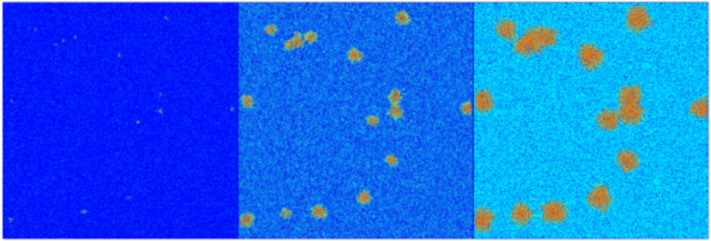


Figure 2. An example to present the epidemic spatial evolution (time (t) = 20, 60, and 120 d).

J Dai, C Zhai, J Ai, J Ma, J Wang and W Sun, "Modelling the Spread of Epidemics based on Cellular Automata", Processes 2021, 9, 55

Modelling forest fire spread

- Each cell represents a region of a forest with a specified tree density
- Transition functions define the trees that are set alight in the next timestep
- Complex transition functions to account for environmental conditions

http://www.macs.hw.ac.uk/~dwcorne/mypages/apps/ca.html

Procedural generation of levels in games

- Represent the level as a grid
- Uses the Moore neighbourhood
- **T** variable used in the transition rule – rock if **T** ≥ 5, floor otherwise

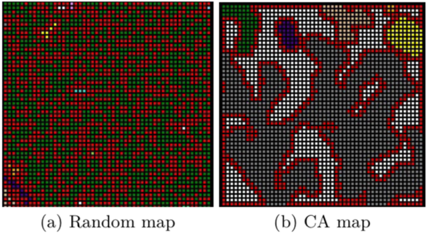


Figure 1: Comparison between a CA and a randomly generated map ($r = 0.5$ in both maps); CA parameters: $n = 4$, $M = 1$, $T = 5$. Rock and wall cells are represented by red and white color respectively. Colored areas represent different tunnels (floor clusters).

L Johnson, GN Yannakakis and J Togelius, "Cellular automata for real-time generation of infinite cave levels", Proc. 2010 Workshop on Procedural Content Generation in Games